

Read the text below. For questions (6—10) choose the correct answer (A, B, C or D). Write your answers on the separate answer sheet.

A BRAIN-BOOSTING VIDEO GAME

In the video game *Tetris*, players try to pack as many shapes as possible into a small space. According to a new study, that's not all they're doing.

It sounds like a joke, but the study uses serious science. A team of three researchers from Canada and the United States scanned the brains of 15 adolescent girls, aged 12 to 15, who played *Tetris*. The scans showed that after 3 months of playing the block-stacking game, grey matter in the girls' brains was thicker. (Grey matter is the wrinkly mixture of brain cells and blood vessels responsible for processing information in the brain.) Part of the thicker grey matter was in a region of the brain near the top of the head. This area, called the parietal lobe, is believed to be responsible for collecting information from the senses.

The study shows that «brain structure is much more dynamic than had been appreciated», says Richard Haier of the University of California, Irvine, one of the three scientists behind the study. Haier says they studied girls' brains because they typically spend less time playing video games than boys.

For comparison, the scientists also scanned the brains of 11 girls who had not been playing *Tetris*. They found no increase in the thickness of those girls' grey matter — suggesting that certain parts of the game-playing girls' brains grew because the girls had played the video game.

The researchers didn't stop there — they also did real-time brain scans of girls while they were playing *Tetris*. For those scans, they used a technique called functional magnetic resonance imaging, or fMRI. An fMRI tracks how blood moves through the brain, and allows scientists to see which brain areas are being used.

These scans showed that in the brains of girls who played *Tetris* for three months, some parts of the brain were being used less. The scientists don't know why. Haier suggests that the drop in activity may be due to the brain actually working more efficiently than before. «We're not sure, but we think the brain is learning which areas not to use», Haier says. «As you learn the game, it becomes more automatic». The parts of the brain that got bigger over the course of three months were not the same parts of the brain that were being used less. This comparison hints that bigger is not always better: just because a part of the brain gets bigger doesn't mean that it's working more efficiently.

«Understanding how the brain works is not easy», says Haier. The scientists don't know if the brain changes due to *Tetris* will help a person learn new skills or have better memory. «We know *Tetris* changes the brain», Haier says. «We don't know if it's good for you».

- 6 Scientists found a connection between playing *Tetris* and _____
- A the thickness of grey matter in the brain.
 - B the size of parts of the brain.
 - C the conditions of the brain cells.
 - D the thickness of the blood vessels inside the brain.
- 7 What does the research of the grey matter show?
- A It was more diluted in the brain near the bottom.
 - B It was responsible for processing blood in the vessels.
 - C It was answerable for collecting information from the senses.
 - D It was thicker in the brain at the top.

- 8 The scientists who were examining the girls' brain knew that boys usually spent _____
 A more time playing video games than girls.
 B less time playing video games than girls.
 C as much time as girls who played video games.
 D not so much time as girls who were accustomed to playing video games.
- 9 The researchers used functional magnetic resonance imaging to _____
 A study how the girls were playing a game.
 B monitor the process of blood circulation through the brain.
 C study the processes of the mental activity.
 D do a scanning of the parts of the brain.
- 10 What does the word «it» in line 24 mean?
 A Grey matter.
 B Brain.
 C Learning.
 D Game.

Task 3

Read the text below. Match choices (A—H) to (11—16). There are two choices you do not need to use. Write your answers on the separate answer sheet.

VITAMINS, MINERALS, AND SUPPLEMENTS

There appear to be a lot of bottles in the supplement area of any store. Every magazine is filled with ads for pills that will change your life, prime-time television has advertisements for a pill for any part of your body and mind, and the Internet has everything for sale.

- 11 Supplements are a multibillion dollar business. The hard questions are: what do we need, how much do we need, and are the ads telling the whole truth? Here is some background information that may help answer the questions. In 1994 Congress passed the Dietary Supplement Act which states that if something is called a dietary supplement, the FDA does not have to test it and no one monitors what is really in the supplement. There has been a lot of abuse since then. Claims are made that may not be true which leaves the consumer at a strong disadvantage. For some strange reason many Americans feel that if something is sold over the counter, or if it says «all natural», that it is safe. Nothing could be further from the truth.
- 12 Many people think, «If a little bit is good, more is better». This thinking can lead to health consequences. So how does one dig through all of the advertising and decide what will enhance their health, what is a waste of money, and what will do no harm? Multivitamins do not fix a lousy diet. Megadose vitamins can be harmful because of having too much of some nutrients. The best bet for a multivitamin mineral tablet is a store brand. High price does not give better quality (about 20 cents a day is the most you should pay). Look for a tablet that provides 100 % of RDA for most nutrients. You cannot get enough vitamin E or calcium in a multivitamin. If you are over age 50, be sure that the tablet has at least 25 mcg of B₁₂. Too much iron is a problem that may be worse than too little iron, so vitamins for mature adults should have no iron.
- 13 Beyond the multivitamin a person may want to supplement vitamin E to get 100—400 IU a day. And, of course, a person wants to supplement the diet to receive a total of 1000—1500 mg calcium depending on age. The only nutrients that have good clinical studies on their benefits are vitamin E, calcium, folate and vitamin D. The folate and vitamin D are in most multivitamins, so vitamin E and calcium are the only supplements needed beyond the multi unless you have been diagnosed through testing by a bona fide physician with a deficiency of a nutrient. Hair analysis, looking in the eyes, etc. are not legitimate tests for deficiency. You need blood tests. In a recent study, one group was given supplements and another group ate better. At the end of the study those taking the multivitamin had little to no change in nutritional status while those on the balanced diet improved their nutritional status. More studies need to be done to confirm this finding.

- 14 In the past few years there has been a significant increase in the number of herbal supplements on the shelf. Some people are using them indiscriminately without realizing that they may have interactions with prescription drugs. Also, studies have been conducted that show that a lot of the herbal supplements do not have enough of the herb in it to be of any value. Remember, there is no governing agency that the manufacturers must answer to. No one is testing for purity or for contents. Herbs have been used in Europe for a long time, but their products are tested for purity and content. Many herbs are considered medicine in European countries.
- 15 Many people take better care of their automobile than they do of their body. They would not dump anything that a friend recommended into the vehicle, but they will do it to their body. We own nothing more precious than our body and keeping it fit and healthy should be our primary concern. Do not live to regret what you did to yourself. Before taking anything, research it carefully looking for valid research by independent researchers from major universities. Do not take something based only on the testimony of somebody else. You would not ask a neighbour to set your broken arm. Do not use them for nutrition information. Talk to the experts in the field, registered dietitians.
- 16 The good news is that the average person is unlikely to take so much of a nutrient that he or she will run into trouble. But it's always wise to consult your doctor before you start using a supplement regularly. And that's definitely true if you're using any supplement in high doses or for prolonged periods of time.

This paragraph dwells on _____

- A** the ill effects of the most popular products.
- B** herbal supplements.
- C** the wrong idea of taking too much vitamins.
- D** the misconception of supplements.
- E** the importance to follow the experts' advice.
- F** food that keeps your body healthy.
- G** clinical studies of energy suppliers usage.
- H** the dosage of vitamins and minerals.

Read the text below. For questions (23—32) choose the correct answer (A, B, C or D). Write your answers on the separate answer sheet.

FARMERS USE FALCONS TO PROTECT BERRIES

Rather than using chemicals or noise to keep birds from eating their ripe berries, some farmers are trying a new way to (23) _____ hungry birds: bigger birds.

Farmers in some parts of the US have (24) _____ people who own falcons, which are large birds of prey, to guard their berries. Small birds called starlings love berries. They can destroy hundreds of thousands of dollars in berries each year. But starlings are afraid (25) _____ falcons, so they fly off when they see one (26) _____. Farmers use falcons that are (27) _____ not to kill the starlings and to fly back to their owners.

Keeping an eye (28) _____ the falcons is hard work. One man and his eight trained falcons (29) _____ last summer at the Roses Berry Farm in Glastonbury, Connecticut. They worked 11 hours a day, seven days a week, to chase starlings (30) _____ the farm.

In the past, the Roses tried chemicals and even small (31) _____ to scare away the birds. But the chemical is no (32) _____ sold and the neighbours did not like the noise from the cannons. The Roses thought about covering their 40 acres of blueberries with nets to protect them, but the netting would have been very expensive.

	A	B	C	D
23	scare off	scare about	scare in	scare on
24	selected	obtained	hired	delegated
25	off	of	from	at
26	nearby	near	nearly	close
27	educated	trained	improved	drilled
28	at	on	in	onto
29	produced	employed	worked	did
30	away	of	from	against
31	flintlocks	munitions	stoves	cannons
32	last	longer	shorter	deeper

Task 6

Read the texts below. For questions (33—42) choose the correct answer (A, B, C or D). Write your answers on the separate answer sheet.

CREATIVE SCULPTURES CAN HELP FEED THE HUNGRY

Did you ever make a tower out of cans? People who participate in a competition called Canstruction get a chance to be (33) _____ with cans of food. At the same time, they are helping to fight hunger.

During Canstruction competitions, teams led by architects, engineers, and builders design and build huge sculptures out of cans of food. More than 50 competitions (34) _____ this year in the US and Canada. Sculptures (35) _____ a shark, a castle, an octopus, a seashell, and even a tornado. Each sculpture (36) _____ between 1,000 and 13,000 cans.

When the Canstruction competition is over, all of the cans of food (37) _____ to food banks, homeless shelters, day care and senior centres, and soup kitchens to help feed hungry people.

	A	B	C	D
33	create	creation	creativity	creative
34	held	are being held	hold	were held
35	has included	include	have included	includes
36	uses	use	using	used
37	is donated	are donated	will donate	have been donated

RETURN TO THE MOON

Would you like to visit the moon? The National Aeronautics and Space Administration (NASA) hopes to land a manned spacecraft on the moon in 2018. That's 46 years after US astronauts last (38) _____ on the moon in 1972.

NASA (39) _____ that the vehicle for the mission to the moon will cost \$104 billion to build over the next ten years.

The new spacecraft will look much different from the winged space shuttle that (40) _____ many astronauts into space in the last 25 years. It will look more like an older spacecraft, called *Apollo*, but it will use some of the new space shuttle technology. Besides going to the moon, the new vehicle will be designed to travel to the International Space Station and to Mars. It will carry four astronauts and more supplies than other spaceships.

At first, astronauts would stay on the moon for about a week. Later, NASA would like to build a base on the moon where crews (41) _____ for six months at a time.

NASA used the *Apollo* rockets for six missions to the moon from 1969 to 1972. The first man to walk on the moon was Neil Armstrong on 20 July 1969. Armstrong and two other astronauts (42) _____ to the moon on *Apollo 11*. Altogether, 12 US astronauts have walked on the moon.

	A	B	C	D
38	walk	walking	walked	walks
39	expects	will expect	expected	had expected
40	carried	have carried	had carried	has carried
41	should work	could work	are to work	will work
42	journeyed	journey	journeys	have journeyed